Lab Skills for Plant Micropropagation

Science

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

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Lab Skills for Plant Micropropagation (A01001)

Short Title:	Lab Skills for Plant Micro		
Faculty	Science and Computing		
Department	Science		
Cluster	Horticulture		
Author	CDALY		
Credits	10	**Level:**	Intermediate

Description of Module / Aims

The module has been designed to enhance the education of science students who will benefit from knowledge of plant culturing techniques, or for students who might require knowledge of establishment and maintenance of environments suitable for sterile culturing of plant, animal or bacterial cultures. Horticultural plant science requires innovation and this module will provide students with the basic requirements to pursue lab-based project work, and eventually level 9 study in plant science.

Programmes

HORT-0041	BSc in Horticulture (WD_SHORT_D)	2 / 3 / E
HORT-0041	BSc in Horticulture (WD_SHORT_D)	3 / 5 / E
HORT-0033	BSc in Horticulture (WD_SHORT_DX)	2 / 3 / E
HORT-0033	BSc in Horticulture (WD_SHORT_DX)	3 / 5 / E

Indicative Content

- Maintaining sterile conditions: practice the principles of maintaining sterile environments and using lab equipment and glassware correctly to maintain sterility and prevent cross contamination
- Commonly used lab calculations: dilutions, molar concentration, commonly used units and measurements and converting between these units
- Lab solutions: preparing and sterilizing nutrient media and other common lab solutions such as phytohormone stock solutions and reagent doses. Budgeting in the lab
- Measuring plant productivity/stress using parameters such as: stomatal frequency, proline content, photosynthetic pigment concentrations and extent of mycorrhizal inoculation
- Lab equipment: calibrate and use of a pH meter and a conductivity meter, and the safe use of centrifuges and autoclaves
- Plant/cyanobacterial cell cultures: components of nutrient media. Establishment and continuous culture of plant callus and liquid cultures. Evaluating the growth rate and viability of cultured plant cells
- Micropropagation of plant clones: propagation of whole plants from sterile ex-plant material
- Cell death in plants: types of cell death in plants including programmed cell death and necrosis and the role of cell death in the plant hypersensitive response. Methods of inducing cell death in plants for scientific study, and enumerating two different types of cell death using staining techniques

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Demonstrate the techniques used to maintain aseptic conditions in a plant lab.
2. Prepare sterile nutrient media and complete an on-going sub-culturing regime for liquid plant and cyanobacterial cell cultures, plant callus and ex-plants.
3. Use and troubleshoot a range of commonplace lab equipment.
4. Calculate the weight of reagents needed to establish solutions of known concentrations and demonstrate proficiency in converting between the units commonly used in the plant science laboratory.
5. Experiment scientifically, interpret the data obtained, appraise observations, and plan further experimentation.
6. Investigate concepts/protocols used in laboratory plant science and evaluate their use.

Learning and Teaching Methods

- Lectures
- Tutorials
- Practical

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	
Lab Report	25%	5
In-Class Assessment	30%	4
Project	25%	6
Practical	20%	1,2,3

Assessment Criteria

<40%: Very limited knowledge and understanding of the subject matter. Lacking in technical ability/ skill and understanding to work effectively in a plant laboratory.

40%-49%: Demonstrate a limited knowledge of the subject matter. Show some technical ability/skill in carrying out and reporting on practical tasks.

50%-59%: Demonstrates satisfactory general knowledge of the main issues within the subject. Student can logically and competently carry out practical tasks.

60%-69%: Demonstrates sound knowledge of the subject matter. Practicals and reports completed to high standard.

70%-100%: Demonstrates excellent technical ability, awareness, and aptitude for working with cultured plant cells. Delivers insightful scientific writing. Develops high competency in practical areas.

Note:

- In addition to the normal regulations, a minimum of 35% must be achieved in both the theory and practical components of the module. In cases where continuous assessment has a practical component attendance of at least 75% in the practicals is mandatory.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	36	
Tutorial	24	
Practical	36	
Independent Learning	174	

Essential Material

- Davey, M.R. and P. Anthony. _Plant Cell Culture: Essential Methods_. Oxford, UK: John Wiley & Son, 2010.

Supplementary Material

- Jones, A., R. Reed and J. Weyers. _Practical Skills in Biology_. 5th ed. Oxford, UK. : Pearson, 2012.
- Lobban, C.S. and M. Schefter. _Successful Laboratory Reports: A Manual for Science S__tudents_. Cambridge, UK. : Cambridge University Press, 1992.

Requested Resources

- Room Type: Biology Lab
- Room Type: Computer Lab